MIL-B-36829A
25 September 1978
SUPERSEDING
MIL-B-36829
4 May 1972

MILITARY SPECIFICATION

BUR, DENTAL, EXCAVATING, ENDODONTIC

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dental excavating steel burs suitable for use with latch type angle handpieces in opening dental pulp chambers for access during endodontic procedures.

1.2 Classification. Dental excavating steel burs shall be of the following shapes and size numbers, as specified (see 6.1):

Shape A. - Round.

Size No. 2
Size No. 4
Size No. 6
Size No. 8

Shape B. - Tapered fissure, crosscut blade.

Size No. 702

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bid or request for proposal form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 6520

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SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods Of.

STANDARDS

FEDERAL

FED-STD-151 - Metals, Test Methods.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 92 - Test for Vickers Hardness of Metallic Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. The burs shall be fabricated from steel having a minimum tungsten content of 3.00 percent, and a minimum carbon content of 1.00 percent. As an alternate, the burs shall be fabricated from steel having a minimum tungsten content of 1.00 percent, a minimum vanadium content of 0.08 percent, and a minimum carbon content of 1.00 percent.

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3.2 Bur shape and dimensions.

3.2.1 Bur shank and overall length of bur. The shape and dimensions of the bur shank and the overall length of the bur shall be as shown in figure 1.

3.2.2 Bur head and neck. Size numbers 2, 4, 6, and 8 burs shall have a round head shape (shape A). Size number 702 bur shall have a tapered fissure crosscut blade head shape (shape B). The dimensions of the bur head and neck shall be in accordance with table I for the applicable shape and size number.

Table I. Head and Neck Dimensions.

Shape	Size number	Number of blades	Head diameter at largest place ± 0.003 inch	Taper or cone angle degrees total 1/	Maximum diameter of neck (inch)	Minimum length of head (inch)
A	2	6	0.039	Not appl.	0.031	0.029
A	4	6	0.055	Not appl.	0.039	0.043
A	6	8	0.071	Not appl.	0.044	0.057
A	8	8	0.091	Not appl.	0.050	0.074
B	702	6 or 8	0.063	4 - 8	0.059	0.169

1/ The total for both sides of the angle.

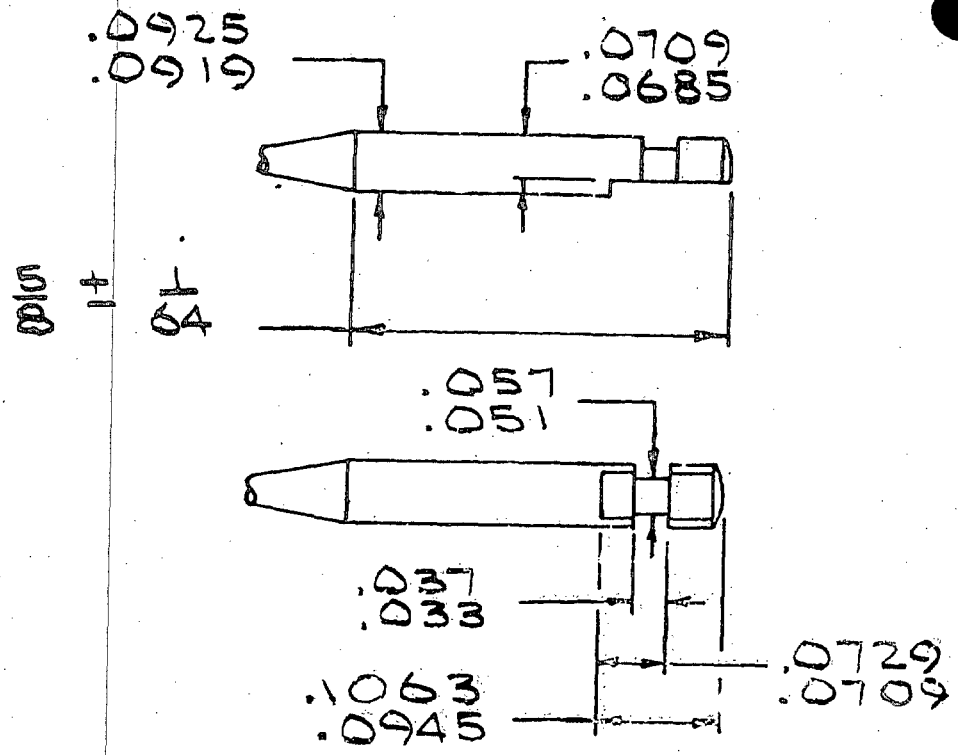
3.3 Design and construction. The burs shall be suitable for use with latch type angle handpieces in opening dental pulp chambers for access during endodontic procedures. Burs shall consist of a shank, a neck, and a head. The blades shall be uniformly cut and sharpened. The burs shall be of one-piece construction.

3.4 Hardness (Vickers test). When tested in accordance with 4.5.2, the hardness of the bur heads shall be no less than number 750 on the Vickers scale (Diamond Pyramid hardness). The bur necks shall be properly tempered to resist fracture or bending when in use.

3.5 Runout. The total indicated runout of the bur shall not exceed 0.003 inch, when tested as specified in 4.5.3.

3.6 Finish. The bur head shall have a finely machined finish, and the blades shall be properly sharpened. The shank and neck shall have a smooth finish.

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NOTE:
ALL DIMENSIONS ARE IN INCHES

FIGURE 1. DIMENSIONS OF SHANK END OF BUR

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3.7 Marking. Each bur shank shall be stamped with the name or trademark of the manufacturer. Such marking shall be legible and permanent.

3.8 Workmanship. Workmanship shall be first class throughout. The burs shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the steel, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered for acceptance are classified as specified in tables II and IV (see 4.2.1 and 4.3) and table III (see 4.2.2).

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4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality (AQL's) shall be as indicated in table II. Unit of product for examination purposes shall be one package (6 burs).

TABLE II. Sampling for Examination.

Type defect	Inspection level	AQL (Percent defective)
Major A	II	1.0
Major B	S-2	1.5
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III. Unit of product for test purposes shall be one package (6 burs).

TABLE III. Tests.

Characteristic	Requirement	Test procedure	Inspection level	AQL (Percent defective)
Material	3.1	4.5.1	1/	2/
Hardness (Vickers)	3.4	4.5.2	S-1	2/
Runout	3.5	4.5.3	S-4	2.5

1/ Not applicable.
2/ Acceptance number shall be zero for all sample sizes.

4.3 Examination. Burs shall be examined for defects including, but not limited to, those listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major A:	
101	Bur head shape or size number not as specified.
102	Burs not latch type (shank) for use with angle handpiece.
103	Style and design not as specified for specific shape and size bur.
104	Blades not properly cut; not sharp and free from nicks, chips, cracks, or pits.
105	Finish of bur heads not as specified.
106	Latch end not cleanly machined.

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TABLE IV. Classification of defects. (Con't)

Categories	Defects
Major B:	
151	Number of blades not as specified for applicable size number of bur.
152	Head diameter not as specified at maximum cross section.
153	Diameter of shank not within specification limits.
154	Width of groove at socket end of shank not within specification limits.
155	Shoulder to shoulder measurement on shank not within specification limits.
156	D-flat dimension on shank not within specification limits.
157	Diameter of socket groove on shank not within specification limits.
Minor A:	
201	Finish of necks and shanks not as specified.
202	Packaging, packing, and marking not as specified.
203	Marking on burs missing or improper.
Minor B:	
251	Overall length of bur not within specification limits.
252	Maximum neck diameter exceeds specification limits.
253	Shank dimensions (other than those specified under Major B defects) not within specification limits.
254	Taper or cone angle of head of No. 702 bur not as specified.

4.3.1 Dimensions. Examination of the dimensions, as listed in 3.2, shall be made with a toolmaker's microscope, or other instrument of equal accuracy. Manufacturer's gages may be used if properly calibrated checking standards are used in conjunction with them.

4.3.2 Microscopic examination. Examination of bur defects not readily perceptible to the naked eye shall be made using a microscope at 10 power (10X) magnification.

4.4 Packaging inspection. The inspection of the unit, intermediate package, packing, and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.5 Tests.

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4.5.1 Chemical analysis of raw material. In the absence of certificates of quality, the chemical composition of the steel used in the manufacture of the burs shall be determined using method No. 111 or 112 of FED-STD-151.

4.5.2 Vickers test. The Vickers hardness test shall be made in accordance with ASTM Designation E 92.

4.5.3 Runout. The total indicated runout of the bur neck shall be measured to the nearest 0.0005 inch. The bur shall be held in a vee-block or precision chuck and rotated slowly. A dial indicator sensitive to 0.0002 inch, equipped with a knife-edge foot or other suitable equipment of equal accuracy, shall be employed. If a dial indicator is used, it shall be mounted with the shaft perpendicular to the axis of the bur and shall be so positioned that the foot contacts the bur neck at the point of smallest diameter. The difference between the maximum and minimum readings for one full rotation of the bur shall be taken as the total indicated runout.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

5.1.1 Unit of issue. One package containing six burs of one shape and size constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Cleaning. Burs shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.2.2 Drying. Burs shall be dried in accordance with MIL-P-116.

5.1.2.3 Preservative application. Preservatives shall not be used.

5.1.2.4 Unit packaging. Six burs of one shape and size only shall be packaged in a plastic or wood holder so designed that each bur is contained in an individual cell. The plastic or wood holder shall be designed with a telescopic, tight fitting, clear plastic cover.

5.1.2.5 Intermediate package. Twelve unit packages containing burs of one shape and size shall be packaged in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification. As an alternate, twelve unit packages may be packaged in a snug fitting clear plastic box having a hinged or plug type closure. The closure shall be completely sealed with a strip of pressure sensitive tape.

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5.1.3 Level C. Six burs of one shape and size shall be packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.1).

5.2.1 Level A. Ninety-six intermediate packages containing burs of one shape and size shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Ninety-six intermediate packages containing burs of one shape and size shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged burs shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking. Each unit package, intermediate package, and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

a. Title, number, and date of this specification.
b. Shape and size required (see 1.2).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Definitions.

(a) Shape. The shape of the bur shall indicate the geometric shape of the bur head.

(b) Size number. The size number of the bur shall indicate both the size and shape of the bur head. For any designated head shape an increase in the bur number shall indicate a larger head and a decrease shall indicate a smaller head.

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6.3 This specification covers the following item(s) appearing in the Federal Supply Catalog:

National Stock Number	Item Identification
6520-00-193-8974	BUR, DENTAL, EXCAVATING, Angle Handpiece, Steel, No. 702, Extra Long Shank, 6s
6520-00-193-8975	BUR, DENTAL, EXCAVATING, Angle Handpiece, Steel, No. 2, Extra Long Shank, 6s
6520-00-193-8976	BUR, DENTAL, EXCAVATING, Angle Handpiece, Steel, No. 4, Extra Long Shank, 6s
6520-00-193-8977	BUR, DENTAL, EXCAVATING, Angle Handpiece, Steel, No. 6, Extra Long Shank, 6s
6520-00-193-8978	BUR, DENTAL, EXCAVATING, Angle Handpiece, Steel, No. 8, Extra Long Shank, 6s

6.4 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activity:
DoD-MB

Preparing activity:
DLA-DM

Project No. 6520-1482

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE
MIL-B-36829A BUR, DENTAL, EXCAVATING, ENDODONTIC

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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